

Yifei YUAN

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EDUCATION

The Chinese University of Hong Kong

Hong Kong

MPhil in Geography and Resource Management

08/2026 – Present

- Supervisor: Prof. Lee Fung, Harry

BSSc in Economics

09/2021 – 07/2026

- GPA: 3.56/4 Major GPA: 3.75/4 **First Class Honours**

Semester Exchange at University of Geneva, Switzerland

02/2024 – 07/2024

- GPA: 5.8/6

RESEARCH INTEREST

My research interests center on the intersection of environmental economics, sustainability assessment, and geospatial technology. I aim to leverage remote sensing applications in social computing, life cycle assessment modeling, and integrated economic-environmental frameworks to evaluate climate policies and emerging carbon neutrality solutions, and to deliver insights for cost-effective and environmentally sound climate action and sustainable development. My current research focuses on the following themes:

- Global-scale assessment of next-generation photovoltaic development;
- Remote sensing applications in social computing;
- Life cycle assessment modeling for sustainability analysis;

PUBLICATIONS

- [1] **Yuan Y**, Sun L, Chen J. Solar poverty alleviation program raises local incomes and lowers air pollution in rural China. *Communications Sustainability*. 2026 **Featured Article** [Paper] [Media]
- [2] **Yuan Y**, Lee HF, Shen Y, Chen J, Sun L, Wang Z. Next-Generation Application of Perovskite Photovoltaics Unlock Global Solar Potential. (Submitted to *Nature Energy*). 2026 [Paper]
- [3] Shen Y, **Yuan Y**, Sun L, Sun G. Photovoltaic as an Effective Solution to Energy Poverty in Developing Countries. (Prepared for submission to *Nature Energy*). 2026 [Paper]

HONOURS & AWARDS

Full Tuition Scholarship & Honours (Top scorer in Gaokao Admission Cohort, Hubei Province)	580,000 HKD
HKSAR Government Scholarship 2022/23 (One of 838 awardees in Hong Kong)	80,000 HKD
HKSAR Government Scholarship 2023/24 (One of 846 awardees in Hong Kong)	80,000 HKD
Yasumoto International Exchange Scholarship (Covers the full cost of study in Switzerland)	20,500 HKD
S.H. Ho Foundation Scholarship (Selected by academic performance and college contributions)	10,000 HKD
S.H. Ho College Advanced Scholarship (Top 5 in 450 based on GPA improvement in 2022/23)	1,500 HKD
Dean's List Award (Top 10%)	Honour
Asia-Pacific Economic Cooperation Scholarship (Awarded by HKSAR with internship opportunities)	Honour
S.H. Ho College Matriculation Scholarships for Academic Excellence (Top 5% at Entrance)	Honour

RESEARCH EXPERIENCE

Chinese Academy of Sciences

Shenzhen

Institute of Technology for Carbon Neutrality, SIAT

12/2024 – 08/2026

Research Intern (Supervisor: Prof. Sun Liqun)

- Leveraged ArcGIS and Python to conduct spatiotemporal analysis of global urban nighttime light dynamics, quantifying significant shifts in urban activity patterns across major regions worldwide.
- Participated in the life-cycle assessment and carbon accounting module for a national-level lithium supply chain governance project. Modeled the full carbon footprint across 5 mining-to-battery production facilities in Jiangxi Province, identifying key emission reduction nodes and delivering data-driven recommendations for policy formulation.

- Built a comprehensive panel dataset by integrating multi-source remote sensing and socio-economic data. Implemented econometric models to identify and quantify the causal effects of environmental policies on regional development.

The Chinese University of Hong Kong

Hong Kong

Department of Economics

09/2025 – 12/2025

Dissertation Project (Supervisor: Prof. Pei Guangyu)

- Built a county-level panel dataset using ArcGIS and Google Earth Engine based on remote sensing data.
- Visualized the energy structure of impoverished counties with Python and constructed PSM-DID models via Stata.
- Empirically verified the significant role of photovoltaic systems in SO₂ emission reduction and revealed the heterogeneous impacts across regions with different income levels and industrial dependence.

CUHK Business School

Hong Kong

School of Accountancy

08/2024 – 09/2024

Research Assistant (Supervisor: Prof. Zhou Yuqing)

- Cleaned and managed economics dataset. Participated in construction of a matched financial dataset of U.S. publicly listed companies and their global subsidiaries, preparing them for analysis and empirical modeling.
- Coordinated with research collaborators and supported research administration tasks, including organizing datasets, literature summaries.

The Chinese University of Hong Kong

Hong Kong

Department of Economics

09/2023 – 12/2023

Individual Project (Supervisor: Dr. Vinci Chow)

- Processed web comment dataset by removing hyperlinks, symbols, and a series of bias factors. Performed word lemmatization and feature engineering to convert nature language to vector.
- Built 5 machine learning classifiers (Logistic Regression, GNB, KNN, Random Forest, SVM), conduct visualization for hyperparameter searching, evaluated model via accuracy/precision/recall/F1, and confirmed SVM as top performers

PROFESSIONAL EXPERIENCE

ByteDance

Shenzhen

Data Analyst Intern

09/2023 – 12/2023

- Conducted research and produced data analysis reports in Southeast Asia market. Provided strategies for operations.
- Participated in e-commerce campaigns and managed weekly operation. Updated strategies based on market report.

EasyTransfer

Beijing

Global Market Intern

05/2023 – 07/2023

- Conducted market research in Hong Kong and produced 2 reports about the market size of Hong Kong.
- Participated in market expansion activity and conducted community operational data analysis.

EXTRACURRICULAR ACTIVITIES

Ronald McDonald's House Charities

09/2025 – 05/2026

- Designed and organized weekend retreat workshop series for sick children and their families, incorporating hands-on craft activities and orientation games to boost engagement and provide emotional support.

Mainland Undergraduate Association Orientation Camp

08/2022 – 10/2022

- Participated in the design of a series of orientation activities to help new students integrate quickly and collected and analyzed feedback to provide insights for activity experience optimization.

SKILLS & INTERESTS

Language: English (fluent), Mandarin (native)

Software: ArcGIS, STATA, Python, JavaScript, OpenLCA

Skill: Global Spatial Analysis, Global Solar Power Simulation, Life Cycle Assessment Modeling, Econometric Modeling